SOMAIYA VIDYAVIHAR UNIVERSITY

22.05.2025 (E)

Semester: January 2025– April 2025			
Maximum Marks: 50	**Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code:01 **Programme:** B.Tech in Computer Engineering	**Class:**	**SY**	**Semester:** IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** COMP		
Course Code: 216U01C402	**Name of the Course:** Analysis of Algorithms		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Solve the following recurrence relation using the Substitution Method and verify using the Master Method. Show all steps clearly and determine the time complexity (asymptotic bound) of the recurrence $T(n) = 3T(n/2) + n^2$	05
ii)	Explain different complexity classes with suitable examples. Comment relationship among P, NP, NP-Hard, and NP-Complete problems.	05
iii)	Solve the following string matching problem using the Longest Common Subsequence (LCS) method: Let: X = "elephant" Y = "relevant" Compute the length of the LCS and also provide the LCS string.	05
Q.2	Attempt any **one**	
i)	a) Discuss the concept of optimal substructure and overlapping sub-problems, solving the Matrix Chain Multiplication problem using the dynamic programming approach. b) Solve the following matrix chain multiplication problem using Dynamic Programming: You are given a chain of 4 matrices with dimensions array as: **P = [6, 4, 2, 4, 5]** Compute the minimum number of scalar multiplications needed to compute the product $A_1A_2A_3A_4$.	10